

Amazon SNS Delivery Policy for HTTP/HTTPS Subscriber

To Begin with the Lab

Summary of the Lab

In this lab, an **HTTP subscriber** was created on an **Amazon EC2** instance using a user-data script, and then an **Amazon SNS Standard Topic** was configured to publish messages to that subscriber. The topic used the default **Delivery Policy**, which retries delivery when an **HTTP/HTTPS** endpoint is temporarily unavailable. The video showed a successful message delivery when the subscriber was ON, then switched the subscriber OFF to simulate a failure and demonstrate SNS retry behavior. After turning the subscriber back ON within the retry window, the same message was delivered successfully, proving that the Delivery Policy controls what SNS does when delivery fails.

Understanding Amazon SQS in a Producer-Consumer Workflow

SNS Delivery Policy tells Amazon SNS what to do when message delivery fails. It mainly applies to **HTTP** and **HTTPS** subscribers, because those endpoints may be temporarily down or return errors. In this lab, the subscriber was hosted on EC2, and the app could be switched ON or OFF to simulate endpoint availability. When the subscriber was OFF, SNS retried the delivery automatically until the endpoint became available again.

Lab Steps

Step 1 – Create the HTTP Subscriber Instance

- Open **Amazon EC2**.
- Click **Launch instance**.
- Enter the instance name: HTTP Subscriber
- Select **Amazon Linux**.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents | My AMIs | **Quick Start**

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI [Info](#) Free tier eligible

Virtualization: hvm ENA enabled: true Root device type: ebs

Summary

Number of instances: [Info](#)

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12... [read more](#)
ami-06067086c1b6c58e6

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as

[Cancel](#) [Launch instance](#) [Preview code](#)

- Choose **t3.micro**.
- Select or create a key pair.
- Use the default VPC.

Instance type [Info](#) [Get advice](#)

Instance type

t3.micro Free tier eligible

Family: t3 - 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand Windows base pricing: 0.0196 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

June-2026-us [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)

vpc-099792461c7d0a616

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply when outside of free tier allowance](#)

Firewall (security groups) [Info](#)

Summary

Number of instances: [Info](#)

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12... [read more](#)
ami-06067086c1b6c58e6

Virtual server type (instance type)

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New security group

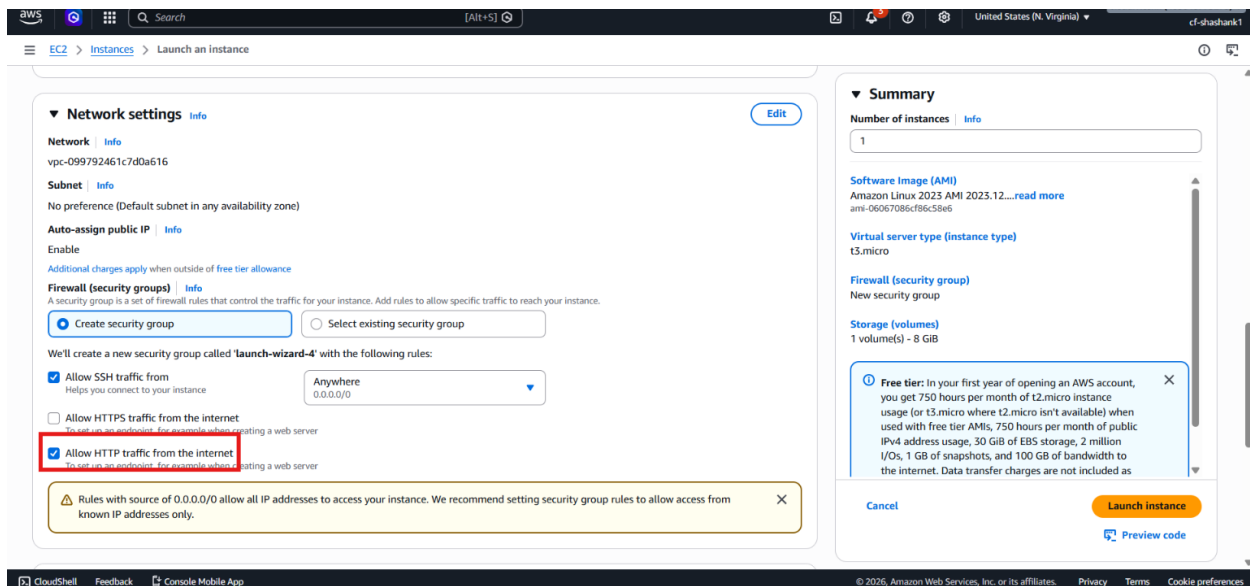
Storage (volumes)

1 volume(s) - 8 GiB

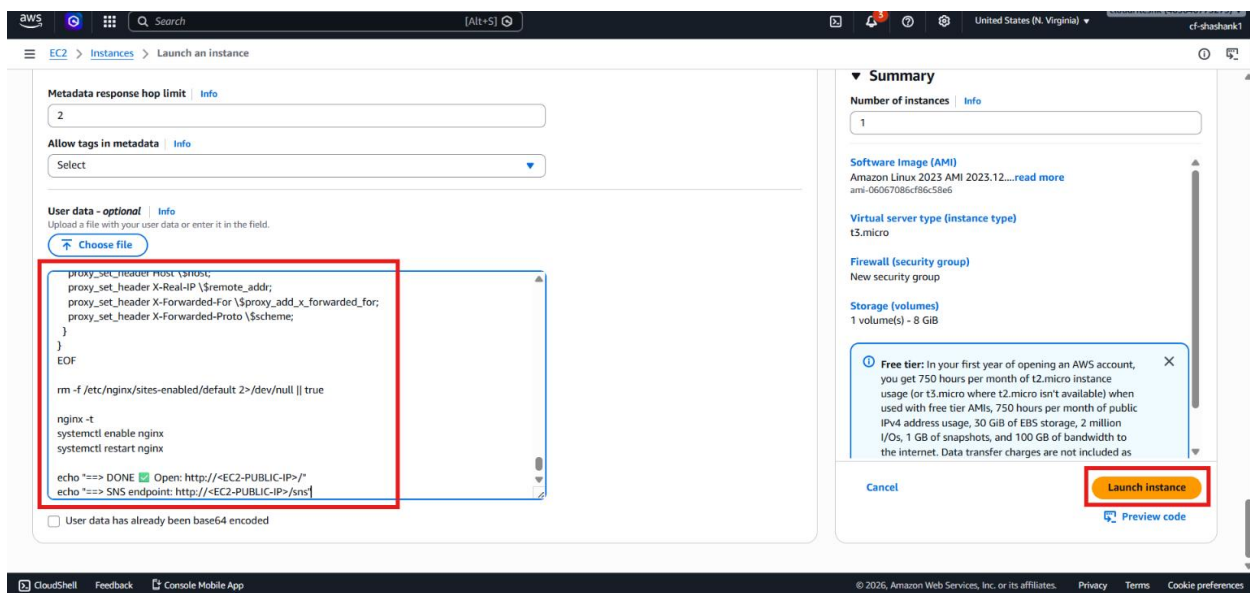
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as

[Cancel](#) [Launch instance](#) [Preview code](#)

- Allow inbound **HTTP** traffic from internet.



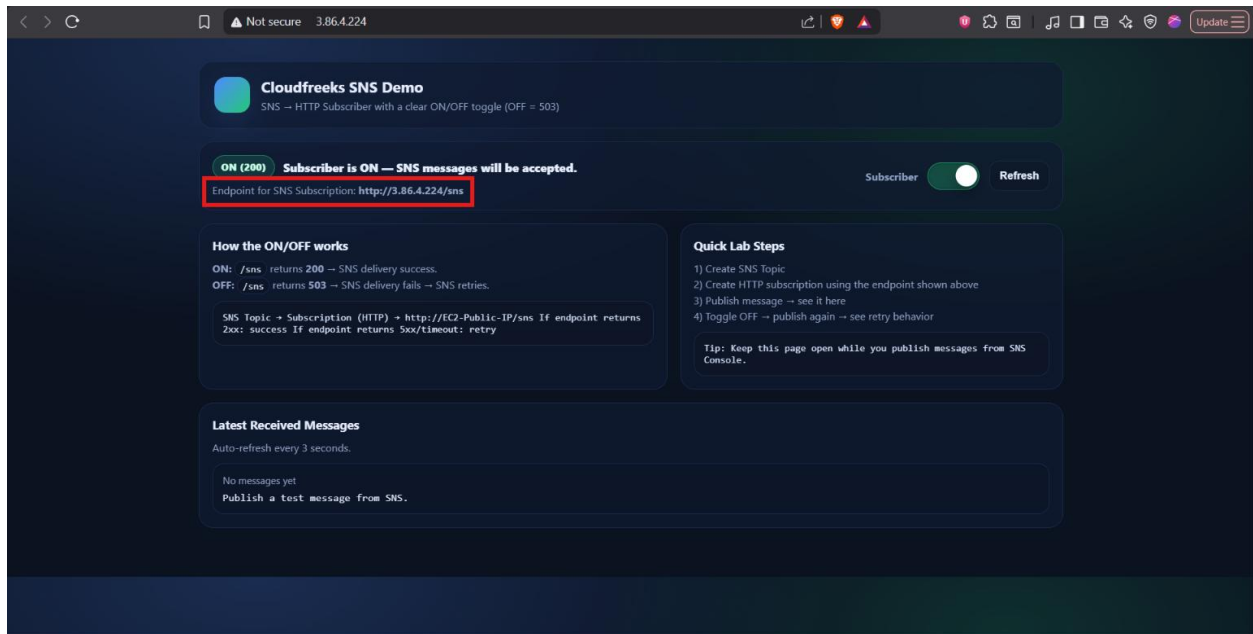
- In advanced details, paste the producer user data script provided in Resources Folder.
- Launch the instance.



- Wait until the web UI is ready.

Step 2 – Open the Subscriber Web UI

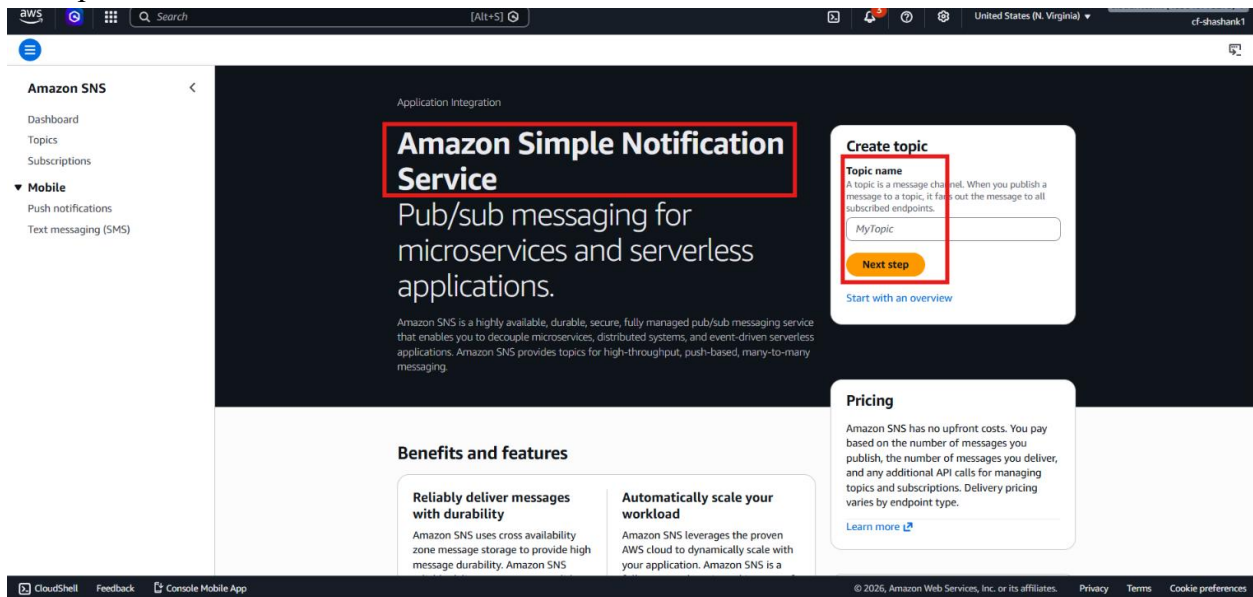
- Copy the EC2 public IP.
- Open it in the browser using **HTTP**.
- Confirm that the subscriber page loads.



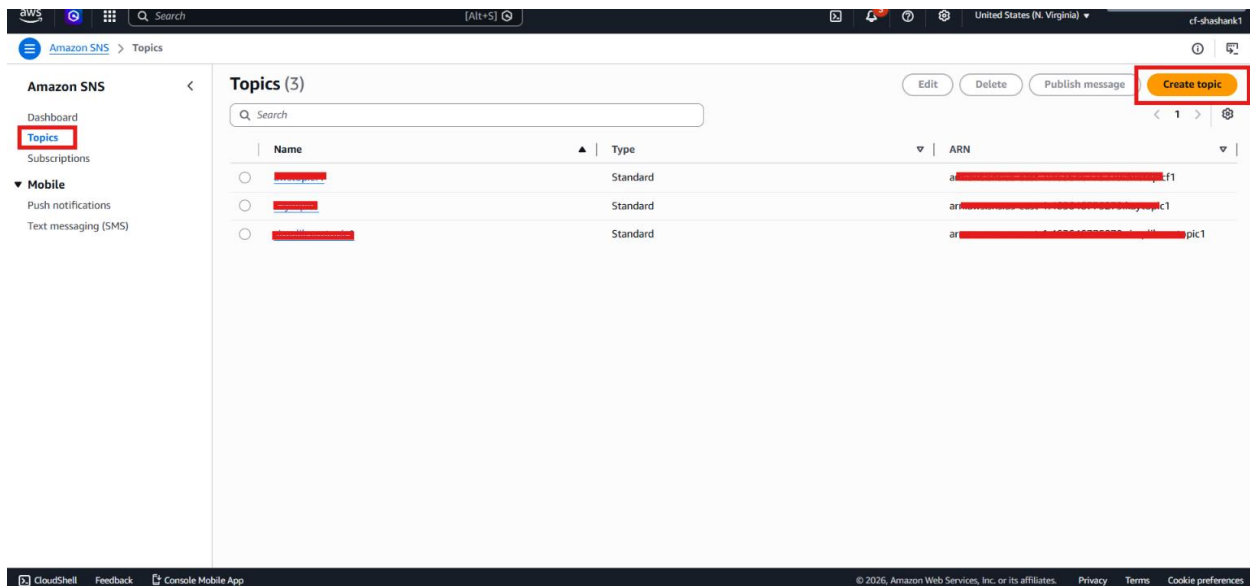
- Keep the subscriber toggle available for testing ON/OFF behavior.

Step 3 – Create the SNS Standard Topic

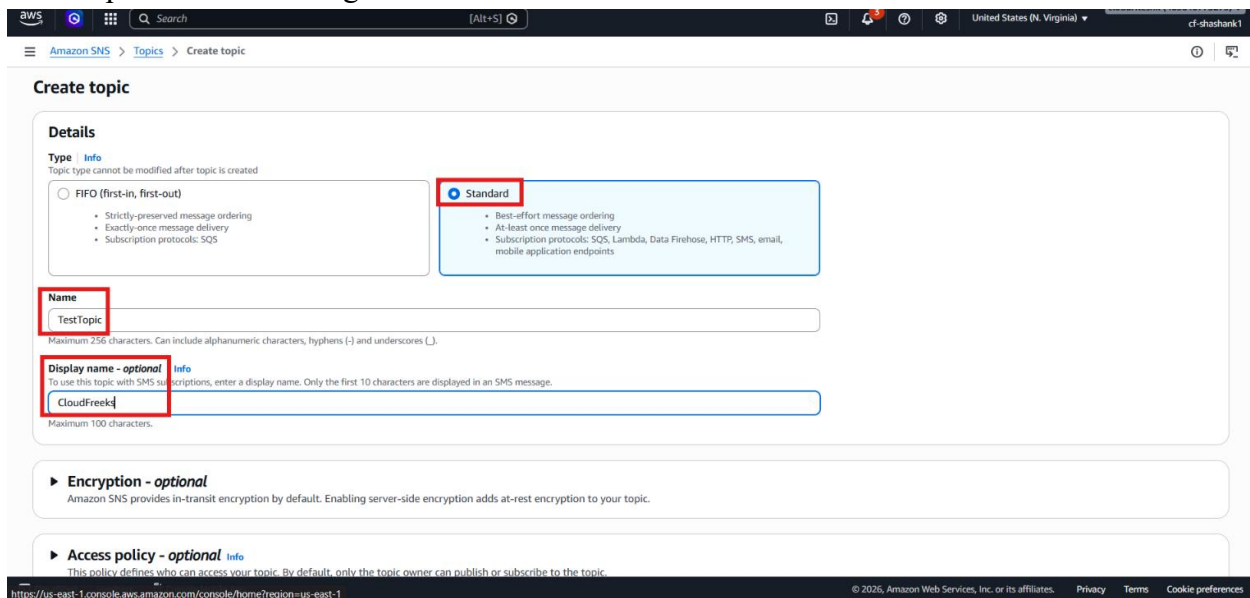
- Sign in to the AWS Management Console.
- Open Amazon SNS.



- Click **Create topic**.



- Select **Standard** topic type.
- Enter the topic name: TestTopic
- Add a display name if required but I will give it as CloudFreeeks.
- Keep the default settings.



- Click **Create topic**.

Step 4 – Review the Delivery Policy in Topic Creation

- Here, expand and review the Delivery Policy section.

The screenshot shows the AWS Management Console interface for creating a new Amazon SNS topic. The 'Delivery policy (HTTP/S) - optional' section is expanded, showing configuration options. The 'Number of retries' is set to 3, and the 'Maximum delay' is set to 20 seconds. A JSON preview is displayed on the right side of the configuration panel.

Delivery policy (HTTP/S) - optional
The policy defines how Amazon SNS retries failed deliveries to HTTP/S endpoints. To modify the default settings, expand this section.

☒ Use the default delivery policy

Number of retries
3

Retries without delay
0

Minimum delay
20 seconds

Maximum delay
20 seconds

Minimum delay retries
0

Maximum delay retries
0

Maximum receive rate
-

Content-Type
text/plain; charset=UTF-8

Retry-backoff function
Linear

JSON preview

```
{
  "http": {
    "defaultHealthyRetryPolicy": {
      "numRetries": 3,
      "numNoDelayRetries": 0,
      "minDelayTarget": 20,
      "maxDelayTarget": 20,
      "numMinDelayRetries": 0,
      "numMaxDelayRetries": 0,
      "backoffFunction": "linear"
    },
    "disableSubscriptionOverrides": false,
    "defaultRequestPolicy": {
      "headerContentType": "text/plain; charset=UTF-8"
    }
  }
}
```

- **Delivery Policy (HTTP/S):** Defines how **Amazon SNS** retries message delivery when an HTTP/HTTPS endpoint is temporarily unavailable.
- **Number of retries (3):** SNS attempts to deliver the message up to **3 additional times** before considering the delivery failed.
- **Maximum delay (20 seconds):** The longest wait time between retry attempts is **20 seconds** (with the default linear backoff).
- **JSON Preview:** Displays the JSON representation of the selected delivery policy settings that SNS will apply to the topic.

Step 5 – Add the HTTP Subscription

- Open the topic.
- Click **Create subscription**.
- Select **HTTP** as the protocol.
- Paste the subscriber endpoint URL from the EC2 web app.

Create subscription

Details

Topic ARN

Protocol
 The type of endpoint to subscribe to.

Endpoint
 A web server that can receive notifications from Amazon SNS.

☐ Enable raw message delivery

After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
 This policy filters the messages that a subscriber receives.

► **Redrive policy (dead-letter queue) - optional** [Info](#)
 Send undeliverable messages to a dead-letter queue.

- Create the subscription.
- Confirm the subscription if needed.

Subscriptions (6)

ID	Endpoint	Status	Protocol	Topic
arn:aws:sns:us-east-1:123456789012:TestTopic	http://3.86.4.224/sns	Confirmed	HTTP	TestTopic
arn:aws:sns:us-east-1:123456789012:TestTopic	http://3.86.4.224/sns	Confirmed	EMAIL	TestTopic
arn:aws:sns:us-east-1:123456789012:TestTopic	http://3.86.4.224/sns	Confirmed	EMAIL	TestTopic
arn:aws:sns:us-east-1:123456789012:TestTopic	http://3.86.4.224/sns	Confirmed	EMAIL	TestTopic
arn:aws:sns:us-east-1:123456789012:TestTopic	http://3.86.4.224/sns	Confirmed	EMAIL	TestTopic
arn:aws:sns:us-east-1:123456789012:TestTopic	http://3.86.4.224/sns	Confirmed	EMAIL	TestTopic

- Verify that the subscription becomes active.

Step 6 – Publish a Test Message

- Open the SNS topic.
- Click **Publish message**.

Amazon SNS > Topics > TestTopic

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

TestTopic

Edit Delete **Publish message**

Details

Name TestTopic

ARN arn:aws:sns:us-east-1:123456789012:testTopic

Display name CloudFreaks

Type Standard

Topic owner 123456789012

Subscriptions Access policy Delivery policy (HTTP/S) Delivery status logging Encryption Tags Integrations

Subscriptions (1)

Edit Delete Request confirmation Confirm subscription **Create subscription**

Search

ID	Endpoint	Status	Protocol
123456789012	http://3.86.4.224/ans	Confirmed	HTTP

- Enter a subject: Test

Amazon SNS > Topics > TestTopic > Publish message

Publish message to topic

Message details

Topic ARN arn:aws:sns:us-east-1:123456789012:testTopic

Subject - optional

Test

Maximum 100 printable ASCII characters

Message group ID - optional

An optional token for SQS fair queue usage. Propagated only to standard SQS queue subscriptions.

Enter message group ID

Maximum 128 characters. Can include alphanumeric characters and punctuation.

Time to Live (TTL) - optional [info](#)

This setting applies only to mobile application endpoints. The number of seconds that the push notification service has to deliver the message to the endpoint.

Message body

Message structure

☒ Identical payload for all delivery protocols. The same payload is sent to endpoints subscribed to the topic, regardless of their delivery protocol.

☐ Custom payload for each delivery protocol. Different payloads are sent to endpoints subscribed to the topic, based on their delivery protocol.

Message body to send to the endpoint

- Enter a message body: Test Message-1
- Publish the message.

Amazon SNS > Topics > TestTopic > Publish message

Message body to send to the endpoint

1 Test Message-1

Message attributes

Type: Select attribute type

Name: Enter attribute name

Value: value or ["value1", "value2"]

Add another attribute

Cancel Publish message

- Confirm that the subscriber receives the message while it is ON.

Cloudfreem SNS Demo

SNS → HTTP Subscriber with a clear ON/OFF toggle (OFF = 503)

ON (200) Subscriber is ON — SNS messages will be accepted.

Endpoint for SNS Subscription: http://3.86.4.224/sns

Subscriber: ☒ Refresh

How the ON/OFF works

ON: /sns returns 200 → SNS delivery success.

OFF: /sns returns 503 → SNS delivery fails → SNS retries.

SNS Topic → Subscription (HTTP) → http://EC2-Public-IP/sns If endpoint returns 2xx: success If endpoint returns 5xx/timeout: retry

Quick Lab Steps

- 1) Create SNS Topic
- 2) Create HTTP subscription using the endpoint shown above
- 3) Publish message → see it here
- 4) Toggle OFF → publish again → see retry behavior

Tip: Keep this page open while you publish messages from SNS Console.

Latest Received Messages

Auto-refresh every 3 seconds.

Time: 2026-07-02T13:46:28.198597Z Type: Notification Topic: **arn:aws:sns:us-east-1:123456789012:TestTopic**

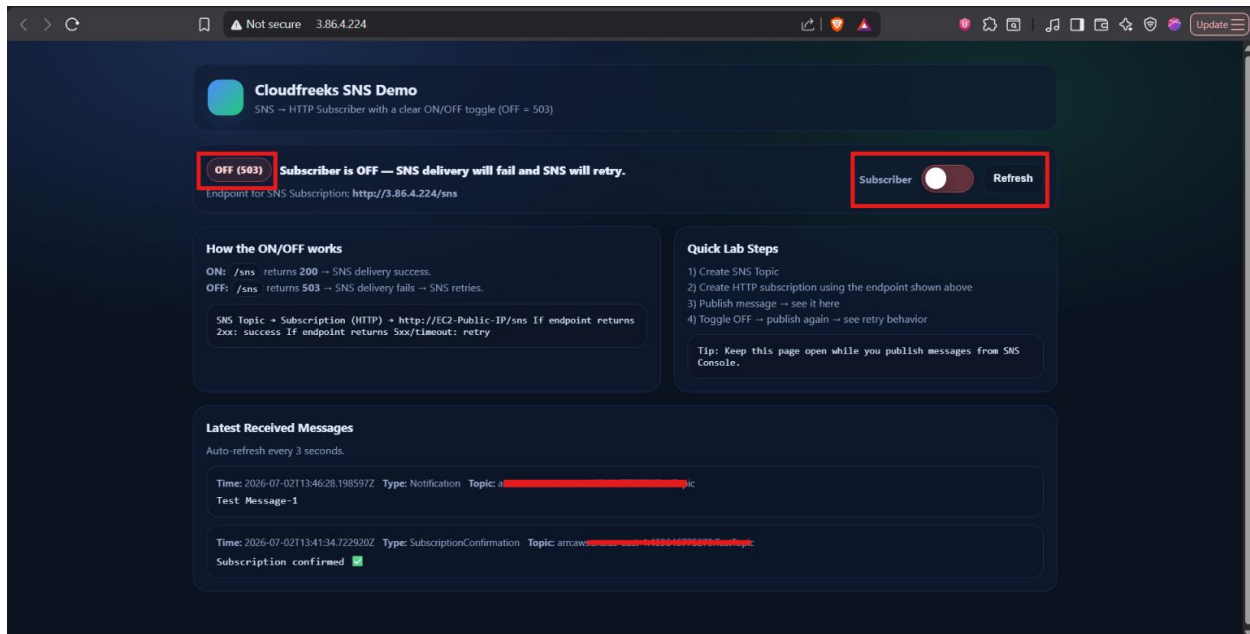
Test Message-1

Time: 2026-07-02T13:41:34.729202Z Type: SubscriptionConfirmation Topic: **arn:aws:sns:us-east-1:123456789012:TestTopic**

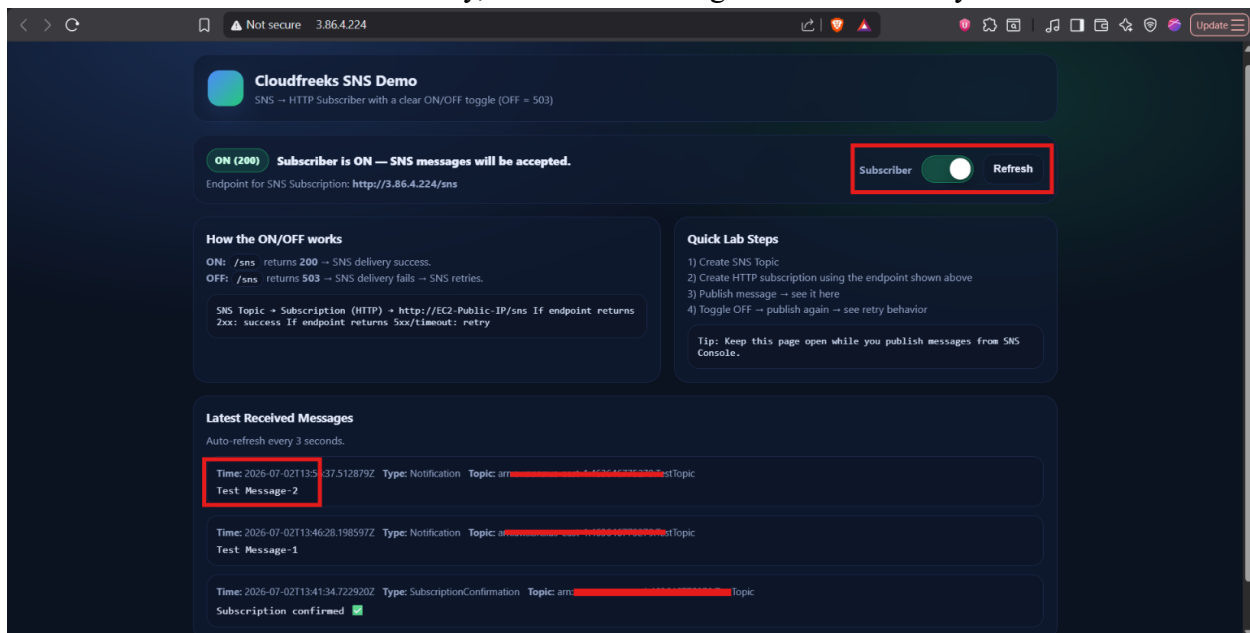
Subscription confirmed

Step 7 – Test Delivery Policy Retry Behavior

- Turn the HTTP subscriber OFF from the web UI.



- Publish another SNS message: Test Message-2
- Observe that delivery fails initially.
- Leave the subscriber OFF briefly, then turn it ON again within the retry window.



- Confirm that SNS retries automatically and the message is eventually delivered.

Verification

- An **HTTP subscriber** was created on EC2.
- An **Amazon SNS Standard Topic** was created successfully.
- The HTTP subscription was added and confirmed.
- Messages were delivered successfully when the subscriber was ON.

- When the subscriber was OFF, SNS retried delivery automatically.
- Turning the subscriber back ON allowed the message to be delivered within the retry period.
- The lab demonstrated how **SNS Delivery Policy** controls failed HTTP/HTTPS delivery behavior.